

Long range electrostatic interactions

Literature Group Meeting

Peter Zhang

04.15.25

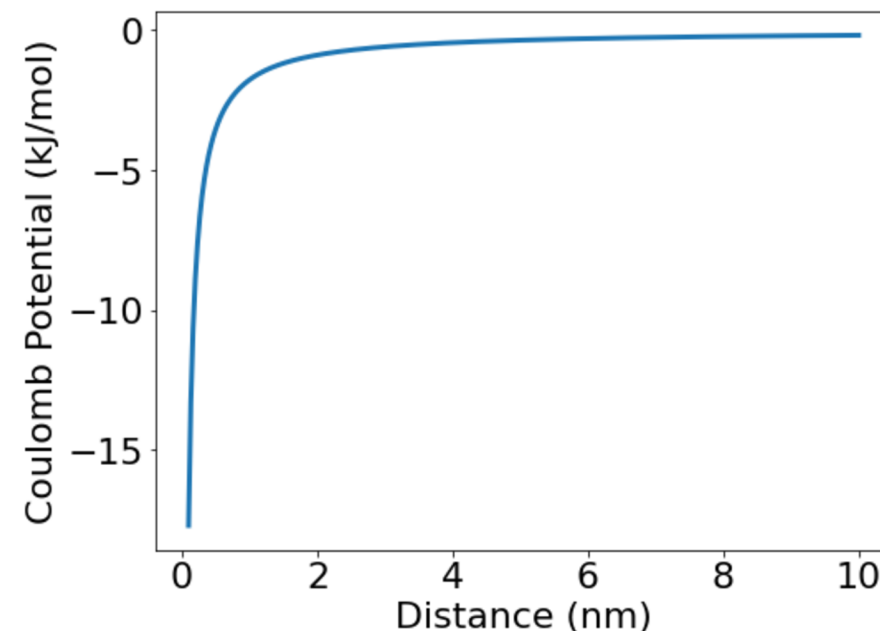
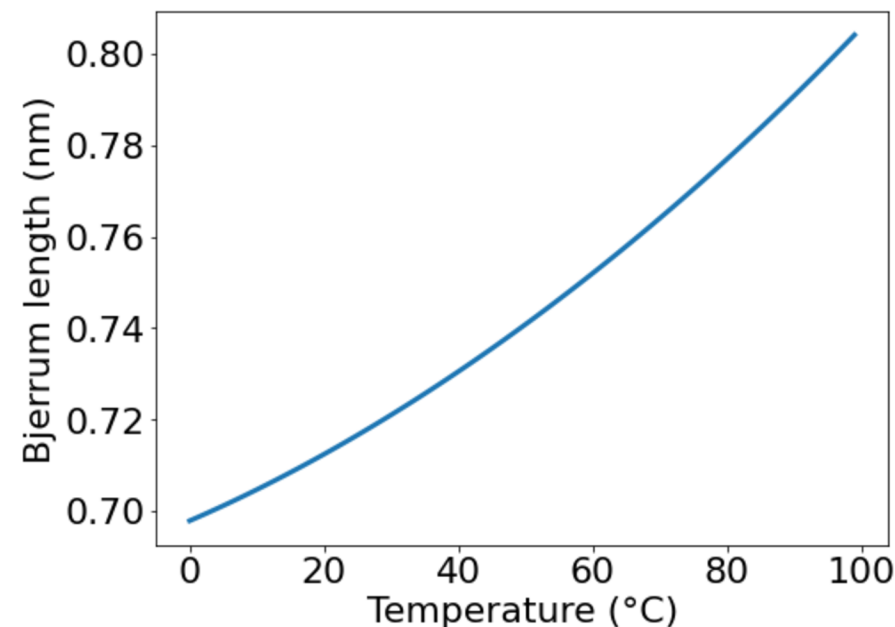
Coulomb energy

Pair energy summation

$$U_c = \frac{l_B}{2} \sum_{i,j=1}^N \frac{q_i q_j}{|r_{ij}|} \quad l_B = \frac{e^2}{4\pi\epsilon_0\epsilon_r k_B T}$$

U_c **Coulomb energy** - the total electrostatic energy (Joules)
In centimetre-gram-second system of units (CGS)
 Electronvolts (**eV**) = 1.602×10^{-19} J

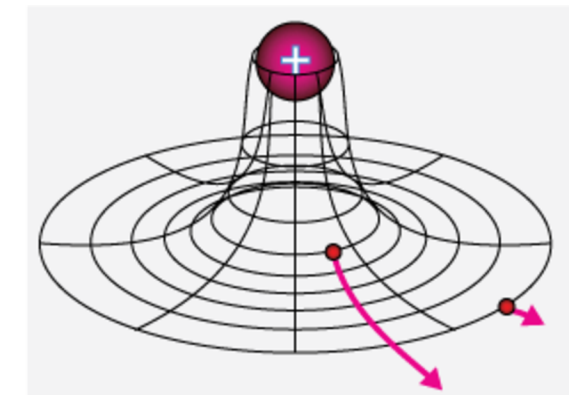
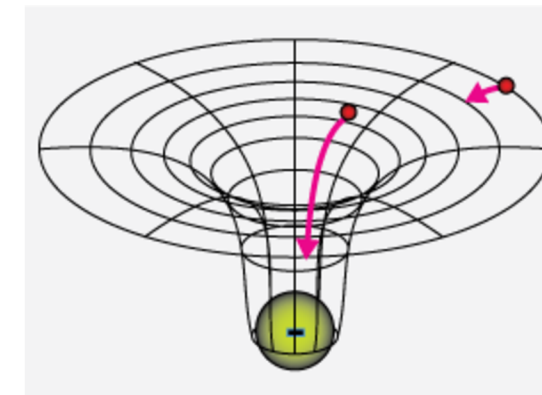
l_B **Bjerrum length** - the distance at which the electrostatic interaction between two point charges is equal to the thermal energy $k_B T$



Potential summation

$$U_c = \frac{1}{2} \sum_{i=1}^N q_i \phi(r_i) \quad \nabla^2 \phi(r) = -4\pi l_B \sum_{j=1}^N q_j \delta(r_j - r)$$

$\phi(r_i)$ **Electrostatic potential at position r_i** - potential energy per unit charge (Joules/Coulomb = Volts)



<https://xaktly.com/Potential.html>

Potential summation derivation

We will define charge distribution using Dirac Delta function:

$$\nabla^2 \phi(r) = -4\pi l_B \sum_{j=1}^N q_j \delta(r_j - r)$$

For single point charge: $\phi(r) = \frac{q_j}{4\pi\epsilon_0\epsilon_r|r - r_j|}$

For a total of N charges: $\phi(r) = \sum_{j=1}^N \frac{q_j}{4\pi\epsilon_0\epsilon_r|r - r_j|}$

Poisson's equation links the Laplacian of Φ to the charge density ρ :

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0\epsilon_r}$$

Note on Laplacian operator ²

$$\nabla^2 \phi = \frac{\partial \phi}{\partial x^2} + \frac{\partial \phi}{\partial y^2} + \frac{\partial \phi}{\partial z^2}$$

The sum of second partial derivatives of Φ in each spatial direction.

$$\rho(r) = \sum_{j=1}^N q_j \delta(r - r_j)$$

$$\phi(r) = \frac{q_j}{4\pi\epsilon_0\epsilon_r} \sum_{j=1}^N \frac{1}{|r - r_j|}$$

$$\nabla^2 \phi(r) = \frac{q_j}{4\pi\epsilon_0\epsilon_r} \sum_{j=1}^N \nabla^2 \left(\frac{1}{|r - r_j|} \right)$$

$$\nabla^2 \left(\frac{1}{|r - r_j|} \right) = -4\pi\delta(r - r_j)$$

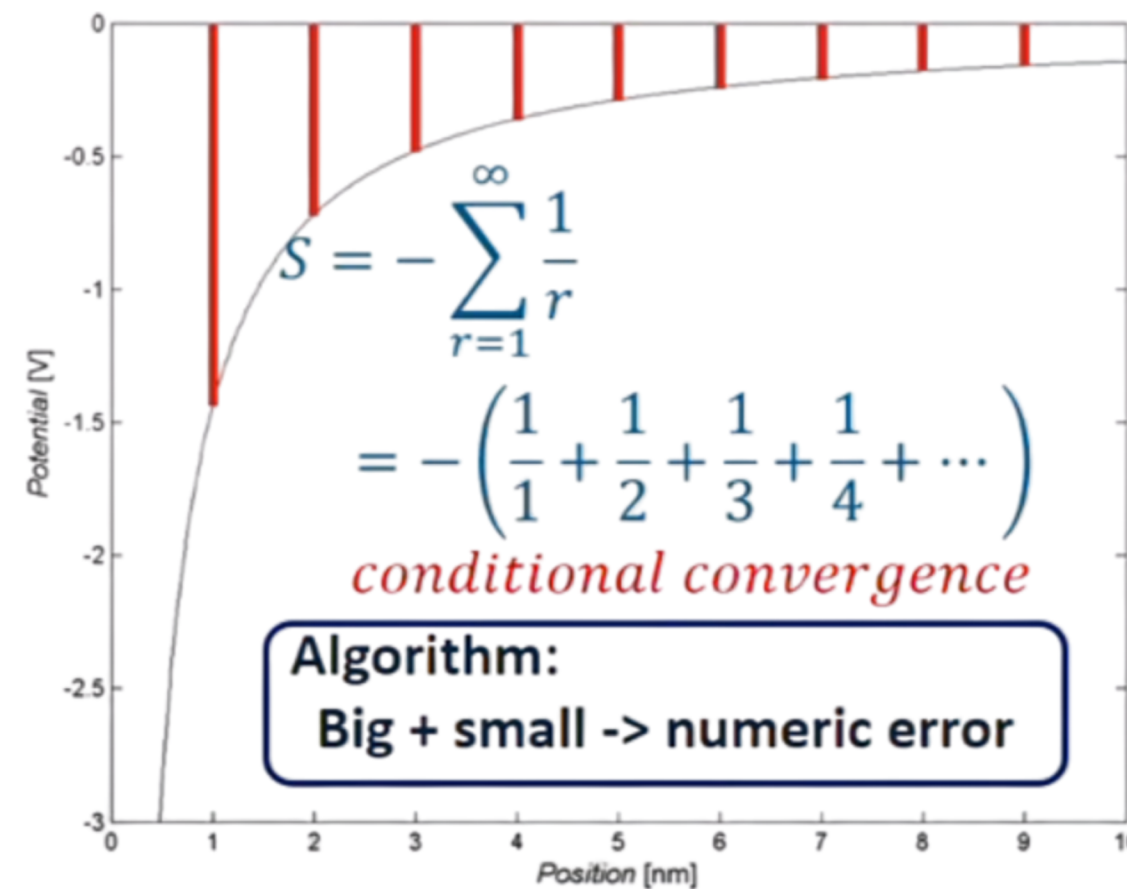
$$\nabla^2 \phi(r) = \frac{1}{\epsilon_0\epsilon_r} \sum_{j=1}^N q_j \delta(r - r_j) = -\frac{\rho}{\epsilon_0\epsilon_r}$$

$$\nabla^2 \phi(r) = -4\pi l_B \sum_{j=1}^N q_j \delta(r_j - r)$$

Electrostatics in Periodic Boundary Conditions

$$\phi(r_i) = \sum'_{j,n} \frac{q_j}{|r_i - (r_j + nL)|}$$

$$U = \frac{1}{2} \sum_{i=1}^N \sum'_n \sum_{j=1}^N \frac{q_i q_j}{|r_i - (r_j + nL)|} = \frac{1}{2} \sum_{i=1}^N q_i \phi(r_i)$$

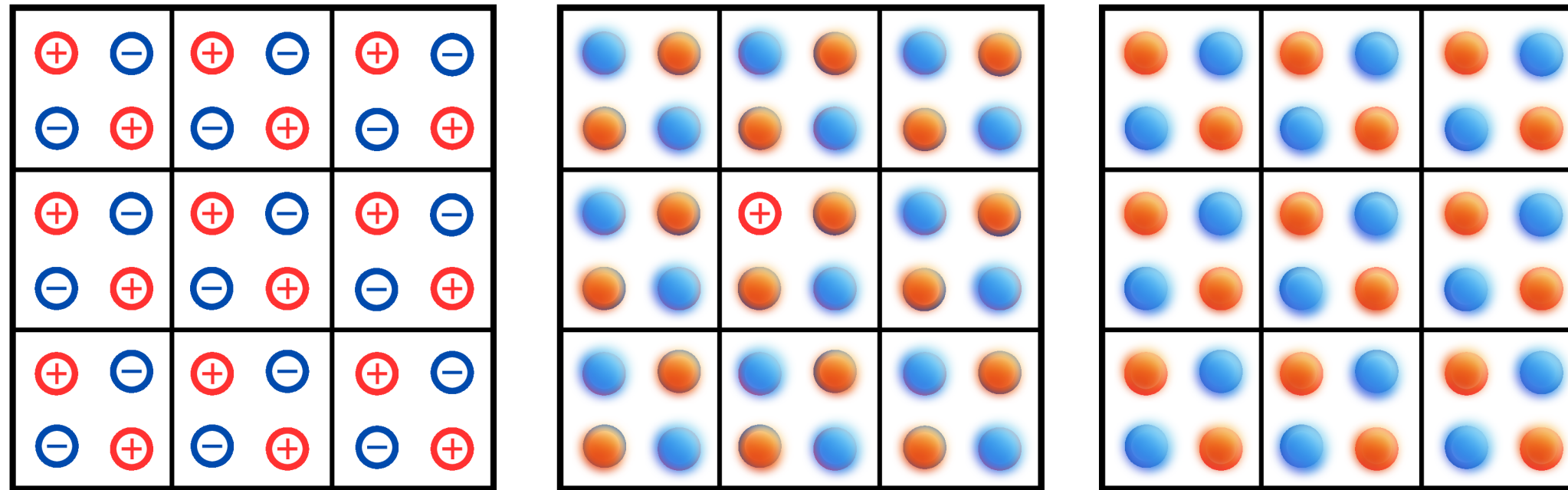


Screening charges

$$U = \frac{1}{2} \sum_{i=1}^N \sum_n' \sum_{j=1}^N \frac{q_i q_j}{|r_i - (r_j + nL)|} = \frac{1}{2} \sum_{i=1}^N q_i \phi(r_i)$$

$$\phi(r_i) = \sum_m' \sum_{j=1}^N \frac{q_j}{|r_i - (r_j + nL)|} = \phi_r(r_i) + \phi_m(r_i) + \phi_0(r_i)$$

$$\rho_{\text{Gauss}}(r) = -q_i (\alpha/\pi)^{\frac{3}{2}} \exp(-\alpha r^2)$$



Fourier Transformation

Consider a periodic system with a cubic box of length L and volume V . Any function $f(r)$ that depends on the coordinates can be represented by a Fourier series:

$$f(r) = \frac{1}{V} \sum_{l=-\infty}^{\infty} \tilde{f}(k) e^{ik \cdot r} \quad k = \frac{2\pi}{L} (l_x, l_y, l_z)$$
$$\tilde{f}(k) = \int_V dr f(r) e^{-ik \cdot r}$$

Laplacian in Fourier space:

$$-\nabla^2 (e^{ik \cdot r}) = k^2 \cdot e^{ik \cdot r}$$

Poisson's equation in real space: $-\nabla^2 \phi(r) = 4\pi \rho(r)$

Poisson's equation in Fourier space: $\phi(r) = \frac{1}{V} \sum_k \tilde{\phi}(k) e^{ik \cdot r}$

$$\rho(r) = \frac{1}{V} \sum_k \tilde{\rho}(k) e^{ik \cdot r}$$

$$-\nabla^2 \phi(r) = -\nabla^2 \left(\frac{1}{V} \sum_k \left(\tilde{\phi}(k) e^{ik \cdot r} \right) \right) = \frac{1}{V} \sum_k k^2 \tilde{\phi}(k) e^{ik \cdot r}$$

$$\frac{1}{V} \sum_k k^2 \tilde{\phi}(k) e^{ik \cdot r} = 4\pi \left(\frac{1}{V} \sum_k \tilde{\rho}(k) e^{ik \cdot r} \right)$$

$$k^2 \tilde{\phi}(k) = 4\pi \tilde{\rho}(k)$$

$$\tilde{\phi}(k) = \frac{4\pi}{k^2} \tilde{\rho}(k) \quad \text{Green's function}$$

Fourier Transformation (cont.)

$$\tilde{\phi}(k) = \frac{4\pi}{k^2} \tilde{\rho}(k)$$

Let's consider point charge z:

In real space: $\rho(k) = z\delta(r)$

In Fourier space: $\tilde{\rho}(k) = \int_V dr z\delta(r)e^{-ikr}$

$$\int dr \delta(r)g(r) = g(0)$$

$$g(0) = ze^{-ikr} = z$$

$$\tilde{\rho}(k) = z$$

Physical meaning: The Fourier transform is a constant - meaning it has constant amplitude across all wave vectors in reciprocal space.

Let's consider smeared charge:

$$\rho_S(r) = \sum_{j=1}^N \sum_n q_j \left(\frac{\alpha}{\pi}\right)^{\frac{3}{2}} \exp(-\alpha^2 |r - (r_j + nL)|^2)$$

$r_j + nL$: center of the Gaussian (R)

$$\rho_S(r) = \sum_{j=1}^N \sum_n q_j \left(\frac{\alpha}{\pi}\right)^{\frac{3}{2}} \exp(-\alpha^2 |r - R|^2)$$

Normalized Gaussian Distribution in 3D

$$G(r) = \left(\frac{\alpha}{\pi}\right)^{\frac{3}{2}} \exp(-\alpha^2 r^2)$$

Fourier Part of Ewald Sum

$$\tilde{\phi}(k) = \frac{4\pi}{k^2} \tilde{\rho}(k)$$

$$\tilde{\rho}_S(k) = \int_V d\mathbf{r} \exp(-i\mathbf{k} \cdot \mathbf{r}) \rho_S(\mathbf{r}) = \sum_{j=1}^N q_j \exp(-i\mathbf{k} \cdot \mathbf{r}_j) \exp\left(-\frac{k^2}{4\alpha}\right)$$

$$\phi_S(k) = \frac{4\pi}{k^2} \sum_{j=1}^N q_j \exp(-i\mathbf{k} \cdot \mathbf{r}_j) \exp\left(-\frac{k^2}{4\alpha}\right)$$

Notice that for a neutral system, the term $\Phi_S(k=0)$ is ill defined:
a direct consequence of the conditional convergence

$$\begin{aligned} U_S &\equiv \frac{1}{2} \sum_i q_i \phi_S(r_i) \\ &= \frac{1}{2} \sum_{k \neq 0} \sum_{j=1}^N \frac{4\pi q_i q_j}{V k^2} \exp[i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)] \exp\left(-\frac{k^2}{4\alpha}\right) \\ &= \frac{1}{2V} \sum_{k \neq 0} \sum_{j=1}^N \frac{4\pi}{k^2} |\tilde{\rho}(k)|^2 \exp\left(-\frac{k^2}{4\alpha}\right) \end{aligned}$$

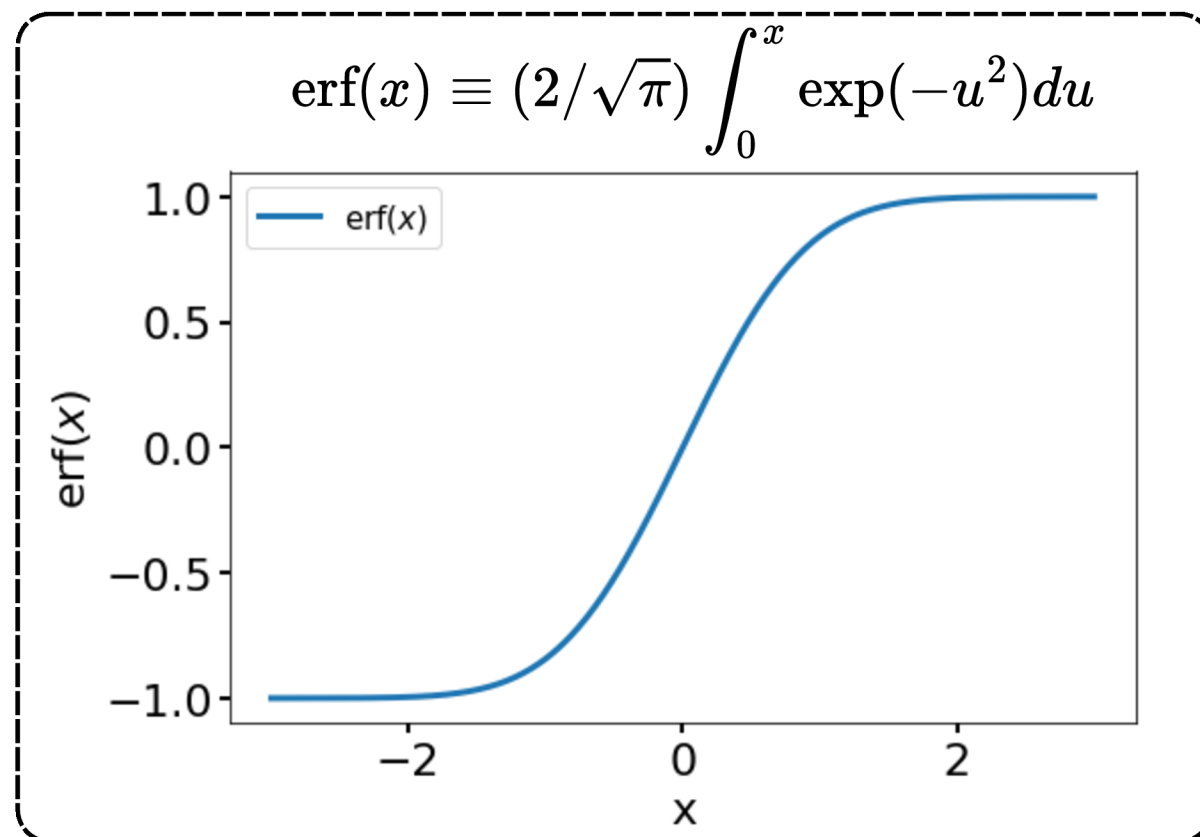
Correction for self-interaction

$$\rho_{\text{Gauss}}(r) = -q_i(\alpha/\pi)^{\frac{3}{2}} \exp(-\alpha r^2)$$

In spherical coordinate: $-\frac{\partial^2 r \phi_{\text{Gauss}}(r)}{\partial r^2} = 4\pi r \rho_{\text{Gauss}}(r)$

First partial integration: $-\frac{\partial^2 r \phi_{\text{Gauss}}(r)}{\partial r^2} = \int_{\infty}^r dr 4\pi \rho_{\text{Gauss}}(r) = 2\pi q_i \left(\frac{\alpha}{\pi}\right)^{\frac{3}{2}} \int_r^{\infty} dr^2 \exp(-\alpha r^2) = 2q_i \left(\frac{\alpha}{\pi}\right)^{\frac{1}{2}} \exp(-\alpha r^2)$

Second partial integration: $r \phi_{\text{Gauss}}(r) = -2q_i \left(\frac{\alpha}{\pi}\right)^{\frac{1}{2}} \int_0^r \exp(-\alpha r^2) = -q_i \text{erf}(\sqrt{\alpha} \cdot r)$



$$\phi_{\text{Gauss}}(r) = -\frac{q_i}{r} \text{erf}(\sqrt{\alpha} \cdot r)$$

$$\phi_{\text{Gauss}}(r=0) = -2q_i \left(\frac{\alpha}{\pi}\right)^{\frac{1}{2}}$$

$$U_{\text{self}} = -\frac{1}{2} \sum_{i=1}^N q_i \phi_{\text{self}}(r_i) = -\left(\frac{\alpha}{\pi}\right)^{\frac{1}{2}} \sum_{i=1}^N q_i^2$$

Real-space sum

Self interaction potential:

$$\phi_{\text{Gauss}}(r) = -\frac{q_i}{r} \text{erf}(\sqrt{\alpha} \cdot r)$$

Real space potential:

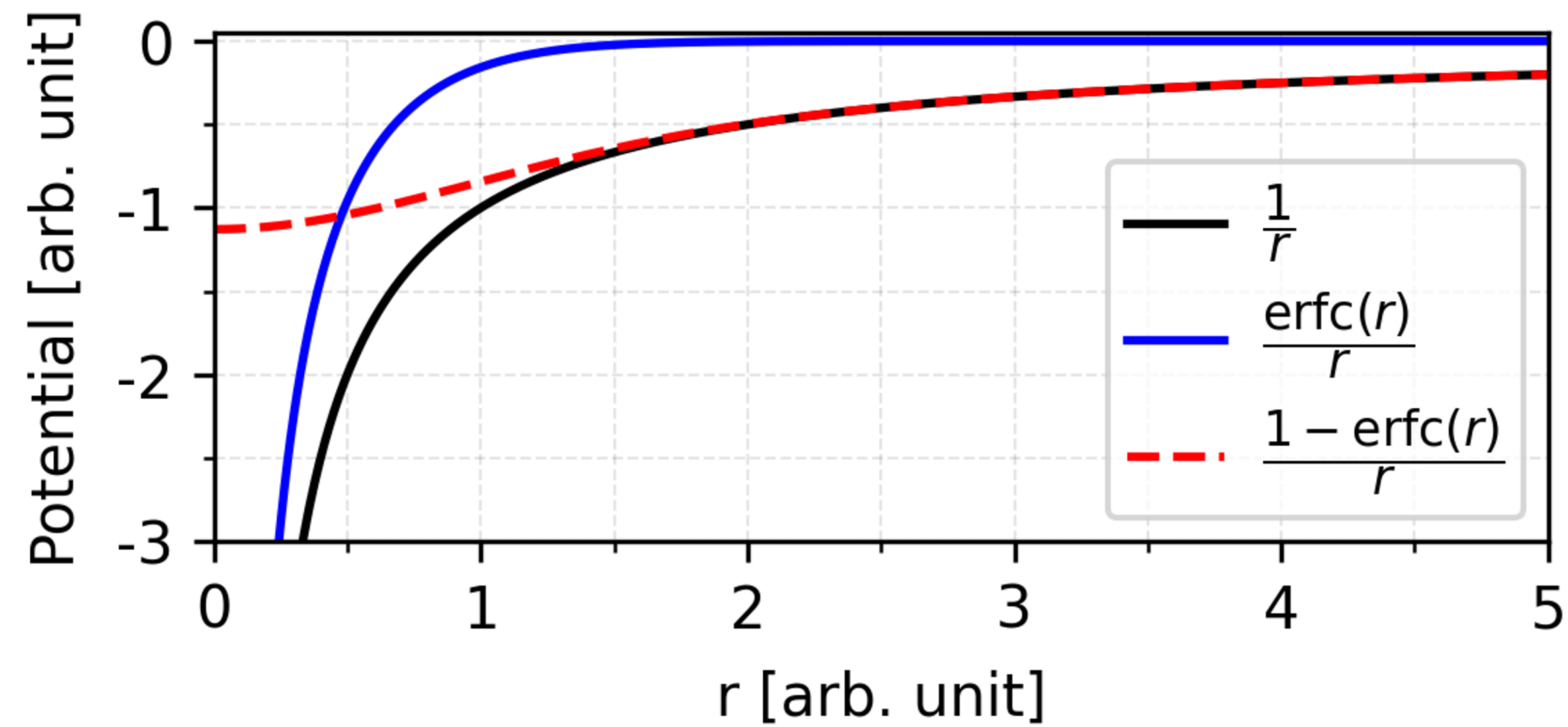
$$\phi_{\text{short-range}}(r) = \frac{q_i}{r} - \frac{q_i}{r} \text{erf}(\sqrt{\alpha} \cdot r) = \frac{q_i}{r} \text{erfc}(\sqrt{\alpha} \cdot r)$$

Real space Coulomb energy:

$$U_{\text{short-range}} = \frac{1}{2} \sum_{i \neq j}^N \frac{q_i q_j \text{erfc}(\sqrt{\alpha} \cdot r_{ij})}{r_{ij}}$$

Ewald Summation

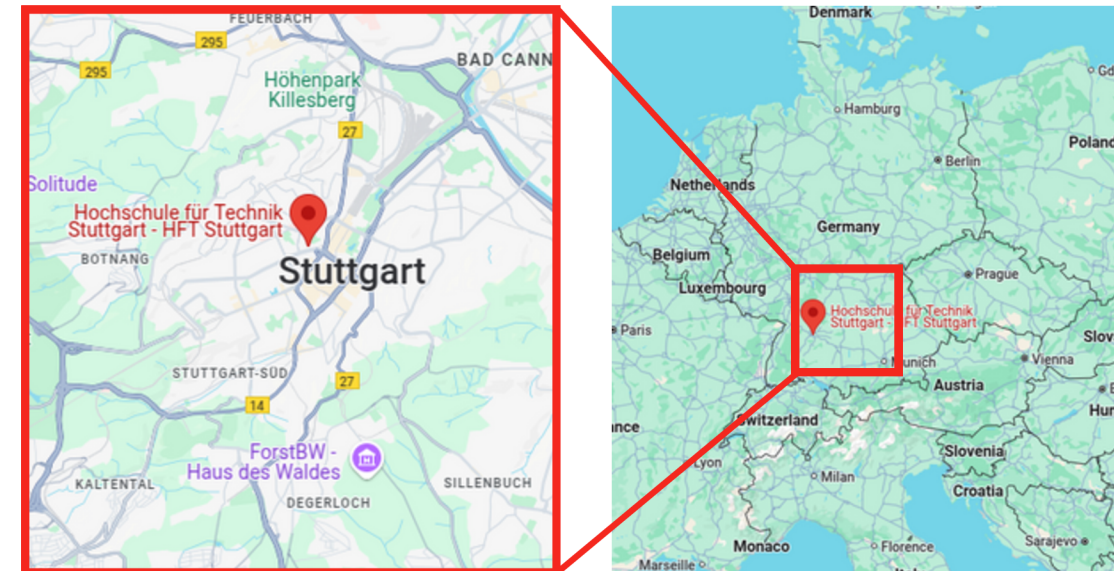
$$U_{Coul} = \underbrace{\frac{1}{2V} \sum_{k \neq 0} \frac{4\pi}{k^2} |\tilde{\rho}(k)|^2 \exp\left(-\frac{k^2}{4\alpha}\right)}_{\text{Reciprocal space}} - \underbrace{\left(\frac{\alpha}{\pi}\right)^{\frac{1}{2}} \sum_{i=1}^N q_i^2}_{\text{Self interaction}} + \underbrace{\frac{1}{2} \sum_{i \neq j}^N \frac{q_i q_j \text{erfc}(\sqrt{\alpha} \cdot r_{ij})}{r_{ij}}}_{\text{Direct space}}$$



Ewald Summation (Cont.)

Paul Peter Ewald (1888 (Munich, Germany) -1985 (Ithaca, New York))

Professor at Stuttgart Technology University of Applied Sciences



3. Die Berechnung optischer und elektrostatischer Gitterpotentiale; von P. P. Ewald.

Inhalt: I. 1. Elektrostatische Potentiale. 2. Elektrodynamische Potentiale. 3. Ziel der Arbeit. — II. 1. Thetafunktionen einer Veränderlichen. Zusammenhang zwischen Thetafunktionen und Gitterproblemen. 2. Ableitung der Transformationsformel für Thetafunktionen von 3 Veränderlichen. — III. 1. Potentiale als Summen von Einzelwirkungen (randloser Kristall). 2. Potentialumformung, Trennungsstelle E . Gesamtpotential eines einfachen Gitters. 3. Erregendes Potential eines einfachen Gitters. 4. Gesamtes und erregendes Potential im zusammengesetzten Gitter, Strukturfaktor. — IV. 1. Übergang zu elektrostatischen Potentialen, dimensionslose Größen. 2. Beispiel: Gitterenergie von Steinsalz. 3. Beispiel: Gitterenergie von Flußspat.

Paul P. Ewald, a Pioneer in Crystal Analysis

Paul P. Ewald, who played a key role in development of the X-ray analysis of crystal structure now widely used in chemistry and physics, died at his home in Ithaca, N.Y., on Aug. 22 after a long illness. He was 97 years old.

His interest in the behavior of light waves within crystals was born while he was working on his doctoral thesis at the University of Munich in 1912. His calculations, initially applicable to visible light, led later to his major contribution: the dynamic theory of X-ray interference in crystals, now used as an analytical tool by solid-state physicists and organic chemists.

At that time, Max Theodor Felix von Laue, then a young instructor at the university, suggested that X-rays impinging on a crystal would be diffracted by atoms in the crystal. This,

he said, would produce a pattern of bright spots dependent on the distribution of atoms in the crystal and the angle of X-ray incidence.

Such an effect could not be achieved with visible light since the wavelength had to be short enough to match the spacing between atoms, but Mr. Laue correctly guessed that X-ray wavelengths were very short. The method was carried further by W. Lawrence Bragg in England and his father, William Henry Bragg.

New Approach Developed

Meanwhile, from 1913 to 1921, Dr. Ewald developed a new way to represent the three-dimensional properties of a crystal under X-ray analysis, the so-called reciprocal lattice approach.

Paul Peter Ewald was born in Berlin

and, after obtaining his doctorate in Munich, worked at several universities, including those at Stuttgart and Paris. In 1932 he was elected rector of the University of Stuttgart.

In 1937, after the rise of the Nazis, he moved to Cambridge University in England, then to Queens University in Belfast, Northern Ireland. Finally, in 1949, he joined the Polytechnic Institute of Brooklyn, now the Polytechnic Institute of New York, as physics professor and department head. In 1959 he became professor emeritus.

Dr. Ewald was the author of several standard texts in crystallography and was active in international scientific efforts. After World War II he helped re-establish the International Union of Pure and Applied Physics and became its secretary general.

While the formulation for X-ray diffraction in crystals developed by the Braggs was simpler than the dynamic theory of Dr. Ewald, a number of physicists believe the Ewald approach provides a deeper insight into the way such rays, and those of visible light, travel through crystals.

Dr. Ewald is survived by his wife, the former Ella Philippson; two sons, Lux and Arnold; two daughters, Rose Bethe and Linde Davidson, and eight grandchildren and two great grandchildren.

Margaret Kennedy, a Leader In Women's Financial Advice

By ETHAN SCHWARTZ

Margaret E. Kennedy, one of the first women to achieve prominence on Wall Street and a leading financial adviser to women, died Monday at St. Vin-



<https://timesmachine.nytimes.com/timesmachine/1985/09/07/issue.html>

1) Vgl. etwa M. Born und A. Landé, Berl. Ber. 45. S. 1048 bis 1088, 1918

Ewald, P. P. *Annalen der Physik* 369, 253–287 (1921).

Applications of Ewald Summation

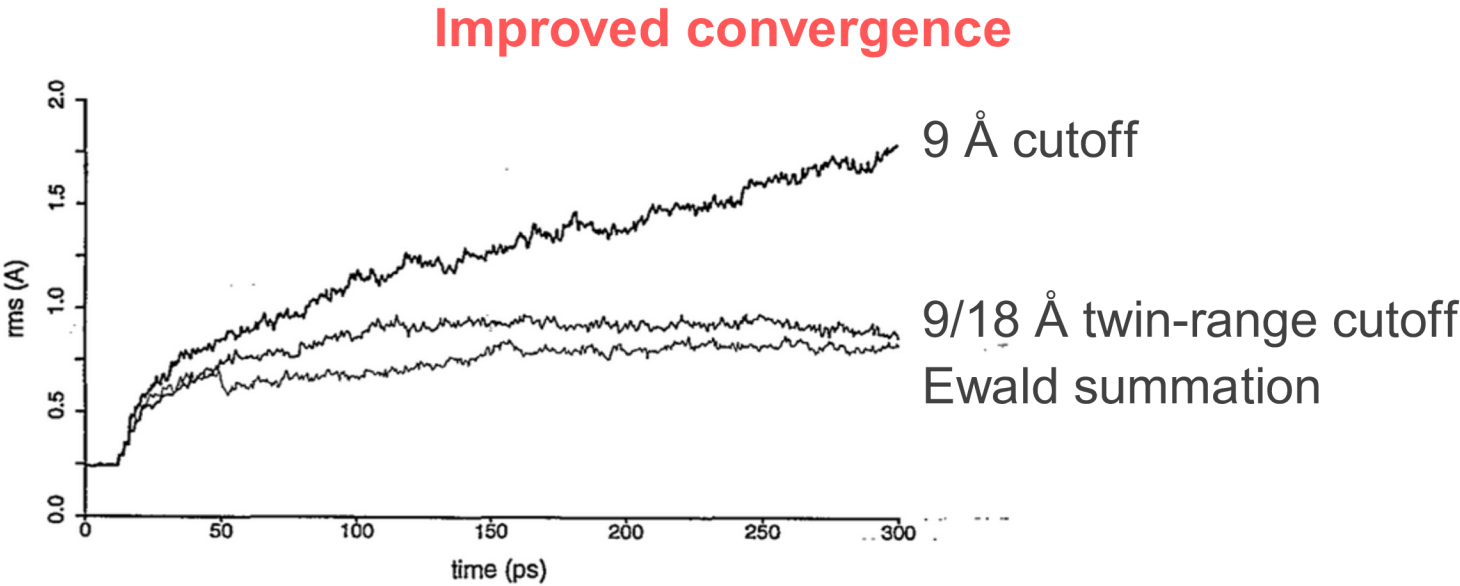
The effect of long-range electrostatic interactions in simulations of macromolecular crystals: A comparison of the Ewald and truncated list methods

York, D. M., Darden, T.A. & Pedersen, L. G. *The Journal of Chemical Physics* **99**, 8345-8348 (1993)

Comparison between cutoff, “twin-range cutoff”, and Ewald summation

Cutoff: Truncating the Coulombic potential (r^{-1}) at a fixed cutoff distance, R_c , so the sum over all pairs is reduced to a sum over a list of non-bonded atoms (the Verlet list).

Twin-range cutoff: Specification of R^{short} and R^{long} . Assume that high-frequency components of the long-range Coulombic forces are negligible.



Improved energy conservation

TABLE II. Energy statistics for the average electrostatic energy $\langle E_{\text{eel}} \rangle$, total energy $\langle E_{\text{tot}} \rangle$, temperature $\langle T \rangle$, and drift in total energy per step $\langle E_{\text{drift}} \rangle$.^a

Simulation	$\langle E_{\text{eel}} \rangle$	$\langle E_{\text{tot}} \rangle$	$\langle T \rangle$	$\langle E_{\text{drift}} \rangle$	CPU ^b
9 Å	-1.33×10^5 (567)	-6.90×10^4 (861)	304.3(5.3)	2.27(1.70)	6.80 ^c
	-1.34×10^5 (390)	-6.91×10^4 (742)	303.9(5.3)	2.78(1.43)	7.68 ^d
9/18 Å	-1.33×10^5 (694)	-6.92×10^4 (1080)	306.4(6.5)	3.88(1.32)	12.50 ^c
	-1.34×10^5 (148)	-7.06×10^4 (220)	297.7(2.0)	0.56(1.57)	29.64 ^d
Ewald	-1.32×10^5 (121)	-6.87×10^4 (18.4)	293.2(1.1)	-0.06(0.004)	15.92 ^c
	-1.32×10^5 (130)	-6.87×10^4 (8.9)	293.4(1.1)	-0.03(0.004)	16.43 ^d

^aConstant energy MD runs (1 ps) restarted at 300 ps; energies are in kcal/mol; rms values are in parentheses.

^bParent processor CPU time for a dedicated 8 processor SGI-380; limited memory on this machine resulted in decreased throughput in the case of the Ewald simulation.

^c20 step nonbond list update.

^d5 step nonbond list update.

Applications of Ewald Summation (cont.)

On the Truncation of Long-Range Electrostatic Interactions in DNA

Jan Norberg and Lennart Nilsson
Center for Structural Biochemistry, Department of Bioscience at Novum, Karolinska Institutet, S-141 57 Huddinge, Sweden
Norberg, J. & Nilsson, L. *Biophys J* **79**, 1537-1553 (2000)

simulation of right-handed double-stranded oligonucleotide d(CGCGCG)2 in B-DNA form with explicit NaCl and TIP3P water

Method	Atom/Group Based	Cutoff distance (Å)	Accuracy (RMSD)	Comment
PME	N/A (Ewald)	N/A	★ Best (~1.5 Å)	Gold standard
AFSH	Atom	12	★ Best (~1.5 Å)	Good PME alternative
AFSHL	Atom	18	★ (~1.6 Å)	Slight improvement
AFSHS	Atom	8	(~1.7 Å)	Slight loss of accuracy
ASH	Atom	12	(~1.9 Å)	Discontinuous forces
AFSW	Atom	8/12	(~2.0 Å)	Smooth but less accurate
ASW	Atom	8/12	(~1.6 Å)	Too rigid
GTRS	Group	8	✗ ~11.8 Å	Catastrophic collapse
GSW / GFSW	Group	8/12	✗ ~4–5 Å	Large deviations
GTR	Group	12	✗ ~4.5 Å	Worse than atom-based

Applications of Ewald Summation (cont.)

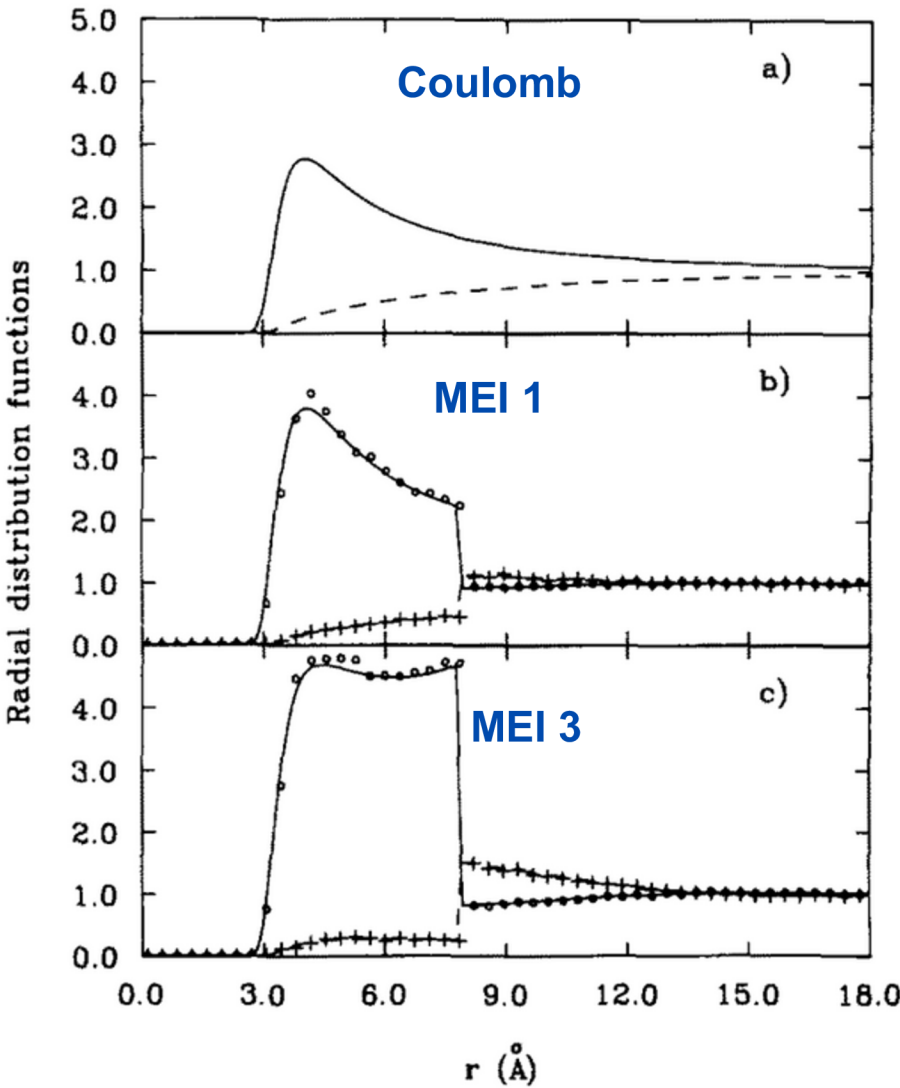
Structural and energetic effects of truncating long ranged interactions in ionic and polar fluids^{a)}

Charles L. Brooks, III,^{b)} B. Montgomery Pettitt,^{c)} and Martin Karplus
Department of Chemistry, Harvard University, Cambridge, Massachusetts 02138

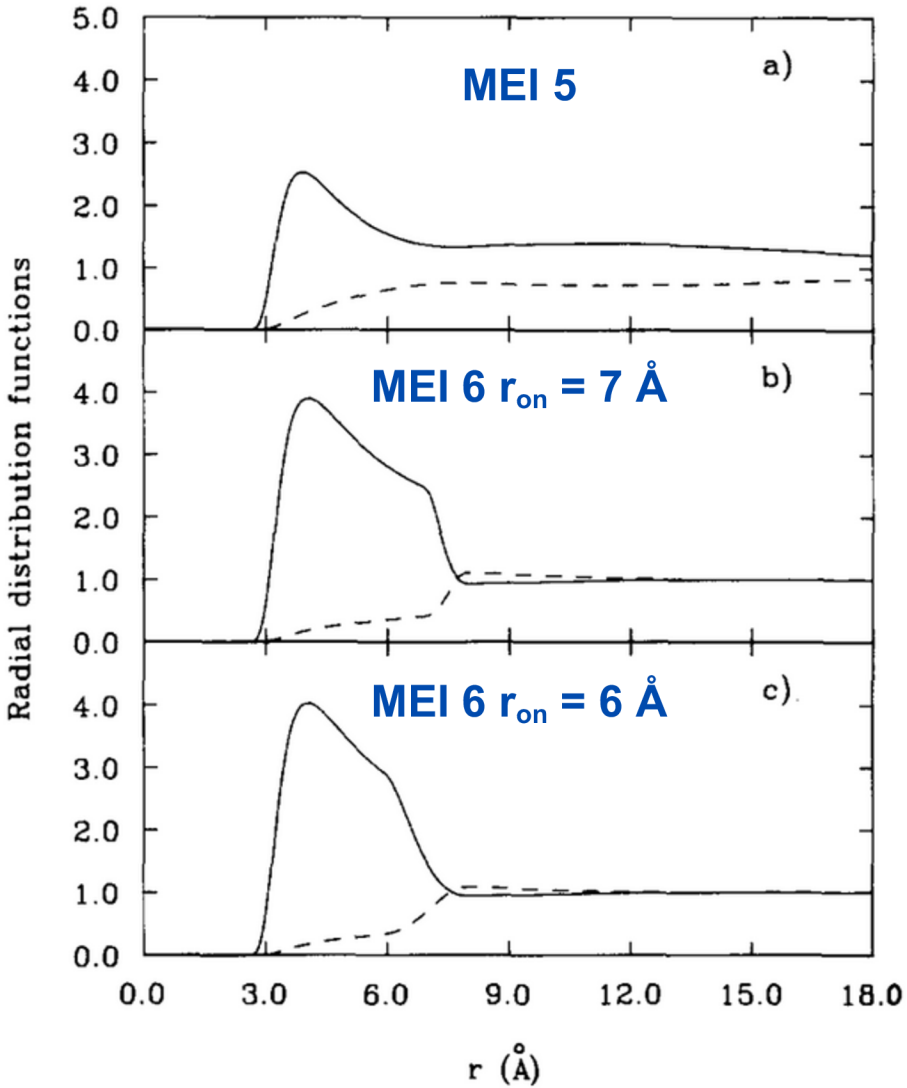
Model system of 0.1 M NaCl in dielectric continuum of dielectric constant 80 and at temperature of 300.0 K

TABLE I. Termination operators for electrostatic interactions.

Model Designation	Operator	Description
MEI1	$S(r) = \begin{cases} 1 & r \leq r_c \\ 0 & r > r_c \end{cases}$	Truncation at r_c
MEI2	$S(r) = \begin{cases} [1 - r/r_c] & r \leq r_c \\ 0 & r > r_c \end{cases}$	Potential zero at r_c
MEI3	$S(r) = \begin{cases} [1 + (r^2/r_c^2)2] & r \leq r_c \\ 0 & r > r_c \end{cases}$	Derivative zero at r_c
MEI4	$S(r) = \begin{cases} [1 - 2(r/r_c) + (r^2/r_c^2)2] & r \leq r_c \\ 0 & r > r_c \end{cases}$	Potential and derivative zero at r_c (lowest order)
MEI5	$S(r) = \begin{cases} [1 - 2(r^2/r_c^2)2 + (r^4/r_c^4)4] & r \leq r_c \\ 0 & r > r_c \end{cases}$	Potential and derivative zero at r_c (higher order)
MEI6	$S(r) = \begin{cases} 1 & r \leq r_{on} \\ \frac{(r_{off} - r)^2(r_{off} - 3r_{on} + 2r)}{(r_{off} - r_{on})^3} & r_{on} < r \leq r_{off} \\ 0 & r > r_{off} \end{cases}$	ST2 switch



Elimination of long-range tail of the interaction without modification of short and intermediate range interaction leads to overemphasis of the correlation functions in 3~6 Å region.



MEI 5 shows reasonable agreement, except a “second” shell of neighbors was seen whereas non exists in the unmodified Coulomb.

Method	Computational Scaling	Advantages	Disadvantages
Direct Coulomb Sum	$\mathcal{O}(N^2)$	Exact (no approximations); simple to implement	Impractical for large systems; very expensive
Cut-off (Truncation)	$\mathcal{O}(N)$	Fast; easy to parallelize	Violates Gauss's law; introduces force discontinuities; artifacts in charged systems
Shifted Potential / Force	$\mathcal{O}(N)$	Smoother than naive truncation; better energy conservation	Still neglects long-range interactions beyond cut-off
Reaction Field (RF)	$\mathcal{O}(N)$	Approximates long-range via dielectric medium; good for homogeneous systems	Inaccurate near interfaces; assumes spherical symmetry and uniform dielectric
Ewald Summation	$\mathcal{O}(N^{3/2})$	Accurate; includes all periodic images	Costly for large systems; not scalable
Particle Mesh Ewald (PME)	$\mathcal{O}(N \log N)$	Gold standard; accurate, scalable; widely available in MD packages	Requires periodic boundary conditions; FFT grid introduces discretization errors
Smooth Particle Mesh Ewald (SPME)	$\mathcal{O}(N \log N)$	More accurate than PME for forces; smoother interpolation	Similar limitations as PME
Fast Multipole Method (FMM)	$\mathcal{O}(N)$	Scalable to millions of particles; no need for periodicity	Complex to implement; less mature in MD packages; performance depends on tuning
Multigrid Methods	$\mathcal{O}(N)$ or $\mathcal{O}(N \log N)$	Efficient for solving Poisson equation; used in some mesh-based codes	Typically only used in mesh/continuum-based MD; not standard for particle simulations
Wolf Summation	$\mathcal{O}(N)$	Energy conserving; smooth decay of interactions; handles neutral systems well	Assumes charge neutrality; accuracy degrades in highly charged or inhomogeneous systems